Mobile Applications Development

Course Number: 020DMOES4/020MADES4
Instructor: Vasken Sarkis

Prerequisites

- Experience in an object-oriented programming language
- Experience in Data Structures
- (Optional) Android device and USB cable

What you'll learn

- How to build Android apps in Kotlin
- Kotlin language essentials
- Best practices for app development

The Duopoly (Market Context)

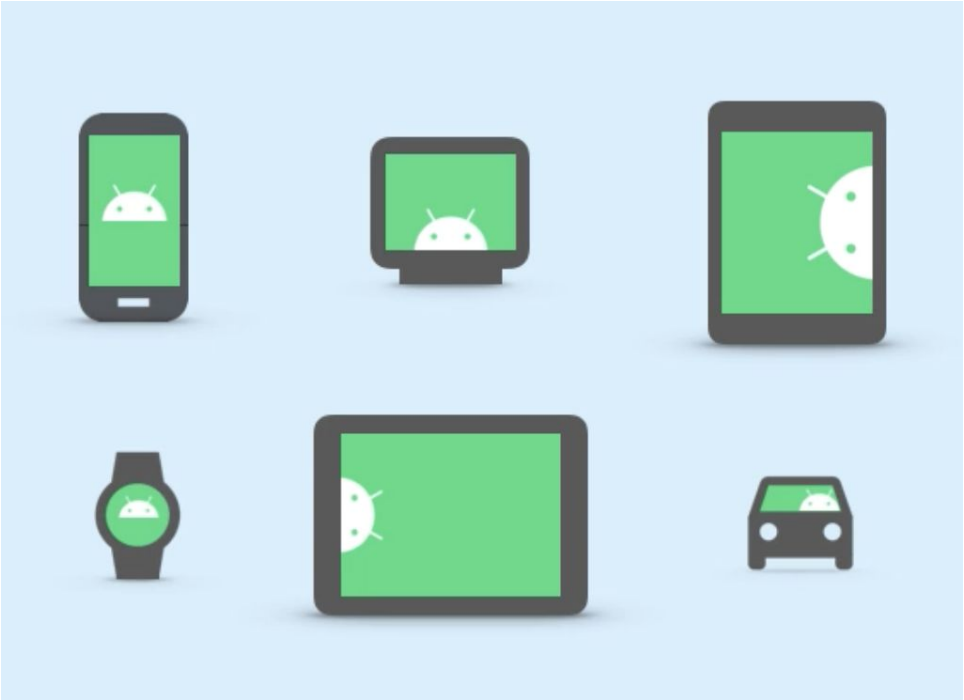
Android:

- ~72% global market share
- Open Source: Runs on phones, watches, cars, tablets, fridges.
- Challenge: Fragmentation (thousands of screen sizes).

iOS:

- ~28% global market share (Dominant in US/Japan)
- Walled Garden: Hardware + Software controlled by Apple.
- Advantage: High user spending & fast OS adoption.

Available across different form factors



Mobile Constraints (The “Big Four”)

- Screen Real Estate:
  - Designing for 5 inches vs. 24 inches. The "Thumb Zone."
- Battery Life
  - The CPU cannot run at 100% continuously. Comparison of "Sleep/Doze" modes vs. Desktop sleep.

Constraints (Continued)

- Connectivity:
  - Handling packet loss, switching from WiFi to 4G/5G, and high latency.
- Input Methods
  - Touch inaccuracies (fat finger problem), voice, gestures, sensors.

Kotlin

A modern programming language that helps developers be more productive.

Benefits of Kotlin

- Expressive and concise
- Safer code
- Interoperable
- Structured Concurrency

MOBILE APP DEVELOPMENT COURSE ROADMAP

Kotlin

Android Studio

WEEK 1: Intro & Kotlin Basics

WEEK 2: Kotlin Continued

WEEK 3: Android First App

WEEK 4: UI & Layouts

WEEK 5: Lifecycle & Navigation

WEEK 6: App Architecture

WEEK 7: Data Persistence

WEEK 8: Networking & APIs

WEEK 9: Advanced Concepts

WEEK 10: Intro to iOS Dev

WEEK 11: Hybrid Approaches

WEEK 12: Capstone Presentations